

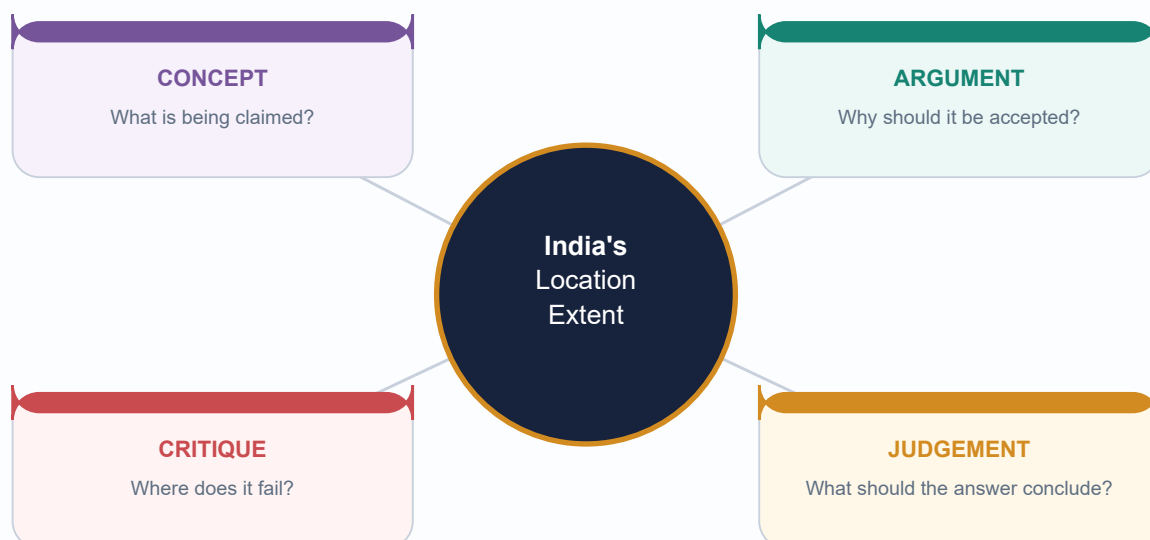
UPSC GEOGRAPHY WORKBOOK

India's Location & Extent Solved-Practice Workbook

Subject: Geography | Paper: GS-I

Eight source-trap MCQs with strict A -> B -> C -> D rotation

Original 10-, 15- and 20-mark solved content frameworks



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

Topic-Specific MCQ Loop 1

GS-I Geography

INTRO & ORIGIN

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q1. Which statement correctly resolves this common misconception: "Earth is a perfect sphere"?

- A. **oblate spheroid/geoid**.

- B. Earth is a perfect sphere

- C. Rotation causes seasons

- D. Latitude fixes time

- ANSWER: A - **oblate spheroid/geoid**.

- Q2. Which statement correctly resolves this common misconception: "Rotation causes seasons"?

- A. Rotation causes seasons

- B. rotation causes day/night; **tilt + revolution** cause seasons.

- C. Latitude fixes time

- D. The whole core is molten

- ANSWER: B - rotation causes day/night; **tilt + revolution** cause seasons.

- Q3. Which statement correctly resolves this common misconception: "Latitude fixes time"?

- A. Latitude fixes time

- B. The whole core is molten

- C. **longitude** fixes time.

- D. Moho separates mantle from core

- ANSWER: C - **longitude** fixes time.

- Q4. Which statement correctly resolves this common misconception: "The whole core is molten"?

- A. The whole core is molten

- B. Moho separates mantle from core

- C. P-waves cannot travel through liquids

- D. **only the outer core** is liquid.

- ANSWER: D - **only the outer core** is liquid.

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE

STUDY LINK

India's Location & Extent -> Source-grounded practice

HIGH

Topic-Specific MCQ Loop 2

GS-I Geography

INTRO & ORIGIN

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q5. Which statement correctly resolves this common misconception: "Moho separates mantle from core"?
 - A. Moho = **crust/mantle**; Gutenberg = mantle/outer-core.
 - B. Moho separates mantle from core
 - C. P-waves cannot travel through liquids
 - D. The Equator passes through India
- ANSWER: A - Moho = **crust/mantle**; Gutenberg = mantle/outer-core.
- Q6. Which statement correctly resolves this common misconception: "P-waves cannot travel through liquids"?
 - A. P-waves cannot travel through liquids
 - B. **S-waves** can't; **P-waves travel through solids and liquids.**
 - C. The Equator passes through India
 - D. India has multiple official time zones
- ANSWER: B - **S-waves** can't; **P-waves travel through solids and liquids.**
- Q7. Which statement correctly resolves this common misconception: "The Equator passes through India"?
 - A. The Equator passes through India
 - B. India has multiple official time zones
 - C. it is the **Tropic of Cancer.**
 - D. Earth is a perfect sphere
- ANSWER: C - it is the **Tropic of Cancer.**
- Q8. Which statement correctly resolves this common misconception: "India has multiple official time zones"?
 - A. India has multiple official time zones
 - B. Earth is a perfect sphere
 - C. Rotation causes seasons
 - D. **single IST** (82°30'E, UTC+5:30).
- ANSWER: D - **single IST** (82°30'E, UTC+5:30).

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE**STUDY LINK**

India's Location & Extent -> Source-grounded practice

HIGH**Solved 10-Mark Practice****GS-I Geography****INTRO & ORIGIN**

Explain India's Location & Extent with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**1. Concept**

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- **Rotation** — spin on axis, ~24 hours (~1,675 kmph at equator) → **day & night**.
- **Revolution** — orbit round the Sun, **365.25 days** → **the year** (leap-year adjustment).
- **Axial tilt** = about **23.5° from the perpendicular to the orbital plane** (66.5° to the plane) → **seasons** and changing day length.
- □ Earth's geometric approximation = **oblate spheroid**; geoid = gravity-defined mean-sea-level surface.
- □ Rotation ≈ 24 h (day-night); Revolution = 365.25 days (year); tilt 23.5° (seasons).
- □ **15° longitude = 1 hour**; IST meridian = **82.5°E (UTC+5:30)**.
- □ International Date Line ≈ **180°** meridian.
- □ Only the **outer core is liquid** (S-wave proof); inner core is solid.
- □ Moho = crust/mantle; Gutenberg = mantle/outer-core.
- Explain how **seismic-wave (P/S) behaviour** maps the Earth's layered, differentiated interior — the
- The longitude–time relationship and the case for/against a **separate North-East India time zone**.
- □ India: **8°4'N–37°6'N** and **68°7'E–97°25'E**.
- □ **Tropic of Cancer (23.5°N)** passes through **8 Indian states**.
- □ IST meridian = **82°30'E** (near Mirzapur); **UTC+5:30**.
- □ ~2-hour sun-time gap between Arunachal and Gujarat, yet a single IST.

MAINS ANGLE

These are model content points; arrange them into paragraphs rather than reproducing a list.

STUDY LINK

India's Location & Extent -> 10-mark answer

HIGH

Solved 15-Mark Practice

GS-I Geography

INTRO & ORIGIN

Analyse India's Location & Extent with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**1. Concept**

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- **Rotation** — spin on axis, ~24 hours (~1,675 kmph at equator) → **day & night**.
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MAINS ANGLE

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STUDY LINK

India's Location & Extent -> 15-mark answer

HIGH

Solved 20-Mark Practice

GS-I Geography

INTRO & ORIGIN

Critically examine India's Location & Extent with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE

1. Concept

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- **Rotation** — spin on axis, ~24 hours (~1,675 kmph at equator) → **day & night**.
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MAINS ANGLE

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STUDY LINK

India's Location & Extent -> 20-mark answer

HIGH**Workbook Rapid Revision****GS-I Geography****INTRO & ORIGIN**

Final revision is placed after teaching and solved practice.

TOPIC RECALL MAP**India's Location & Extent****V**

Definition Rotation — spin on axis, ~24 hours (~1,675 kmph at equator) → day & night .	Mechanism Revolution — orbit round the Sun, 365.25 days → the year (leap-year adjustment).
Application Axial tilt = about 23.5° from the perpendicular to the orbital plane (66.5° to the plane) → seasons and changing day length.	Evaluation ☐ Earth's geometric approximation = oblate spheroid ; geoid = gravity-defined mean-sea-level surface.

STATIC THEORY

- **Rotation** — spin on axis, ~24 hours (~1,675 kmph at equator) → **day & night**.
- **Revolution** — orbit round the Sun, **365.25 days** → **the year** (leap-year adjustment).
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- ☐ Earth's geometric approximation = **oblate spheroid**; geoid = gravity-defined mean-sea-level surface.
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- ☐ **15° longitude = 1 hour**; IST meridian = **82.5°E (UTC+5:30)**.
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- ☐ Only the **outer core is liquid** (S-wave proof); inner core is solid.
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- Explain how **seismic-wave (P/S) behaviour** maps the Earth's layered, differentiated interior — the
- The longitude–time relationship and the case for/against a **separate North-East India time zone**.
- ☐ India: **8°4'N–37°6'N** and **68°7'E–97°25'E**.

MEMORY HOOK

Meaning -> mechanism -> evidence -> limitation -> judgement

MUST-KNOW FACTS

1. ☐ **Tropic of Cancer (23.5°N)** passes through **8 Indian states**.
2. ☐ IST meridian = **82°30'E** (near Mirzapur); **UTC+5:30**.
3. ☐ ~2-hour sun-time gap between Arunachal and Gujarat, yet a single IST.
4. ☐ India's officially remeasured coastline is **11,098.81 km**. The legacy
5. Use India's longitudinal extent to argue **for/against a separate North-East time zone** — energy and
6. How India's latitudinal spread (tropical to sub-tropical) shapes its climatic diversity.

UPSC TRAPS

WRONG: Earth is a perfect sphere

CORRECT: oblate spheroid/geoid.

WRONG: Rotation causes seasons

CORRECT: rotation causes day/night; **tilt + revolution** cause seasons.

WRONG: Latitude fixes time

CORRECT: longitude fixes time.

WRONG: The whole core is molten

CORRECT: only the outer core is liquid.

MAINS ANGLE

Use India's Location & Extent as an argument, not as an unstructured list of facts.

STUDY LINK

Geography -> India's Location & Extent

End of Register Notes - UPSC Agent / Copilot CLI